

25AID432 · 50% OF THE COURSE

Class Assignment: The Qubit Workbook

Three parts, one for each unit, and a five-minute viva. You work the mathematics by hand, check it in NumPy, and finish by writing a small qubit simulator and using it.

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COURSE

25AID432 Mathematical Postulates of Quantum Mechanics

WEIGHT

50% of the course

MARKS

50

WORK

Individual

DEADLINES

Set on Classroom 50 and announced in class

SUBMIT

Through Classroom 50, classroom qts-mpqt

OFFERING

Odd Semester 2026-27

What this is

Half of your grade in this course comes from one workbook, built up across the semester in three parts, one per unit of the course. Each part is submitted through Classroom 50, and all three close near the end of the semester, before the viva. A short viva at the end checks that the work is yours.

PART	WHAT YOU DO	MARKS	CLASSROOM 50 ITEM
A, complex numbers	8 problems by hand, plus a NumPy check file	10	<code>mpqt-a</code>
B, linear algebra	9 problems by hand, plus a NumPy check file	14	<code>mpqt-b</code>
C, a qubit simulator	Write <code>qubit.py</code> (autograded), then a 2–3 page report of three experiments	20	<code>mpqt-c</code>
Viva	Five minutes: explain your own code and your own working	6	In class, after Part C closes
Total		50	

The deadline of each part is shown on Classroom 50 and announced in class. Do each part as its unit finishes; leaving all three to the last fortnight is the surest way to lose marks.

Parts A and B are the same kind of arithmetic you do in the session examples and in the self-assessment questions of the notes. Part C turns the postulates of Unit 3 into working code. The three parts build on each other: the inner product you practise by hand in Part B is the `inner` function you write in Part C, and the relative-phase problem that closes Part A is the interference experiment in Part C.

How to present handwritten work

Parts A and B are worked by hand on paper, on purpose. This arithmetic is learned with a pen, and the end-semester examination is written on paper.

- Number every problem and every sub-part as on this sheet, and keep them in order.
- Show every step. A bare answer earns at most the final mark of a problem; the method earns the rest.
- Box or underline each final answer.
- Scan all pages, in order, into **one PDF** with any phone scanner app. Check the scan is readable before you submit. Unreadable work is marked as missing.

Part A: complex numbers (10 marks)

Covers sessions 1 to 3. Accept the Classroom 50 item `mpqt-a` and commit `partA.pdf` (the scan) and `checks_a.py` (the NumPy checks).

Throughout, $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$.

A1. Cartesian arithmetic (1 mark). Compute, in the form $a + bi$:

- (a) $z_1 + z_2$ (b) $2z_1 - z_2$ (c) $z_1 z_2$ (d) z_1^2

A2. Conjugate and modulus (1 mark). Let $z = a + bi$ with a and b real.

- (a) Show that $zz^* = a^2 + b^2$, and hence that $|z| = \sqrt{a^2 + b^2}$.
- (b) Show that $z + z^* = 2 \operatorname{Re}(z)$ and $z - z^* = 2i \operatorname{Im}(z)$.
- (c) Using your answer to A1(c), check by hand that $(z_1 z_2)^* = z_1^* z_2^*$ and that $|z_1 z_2| = |z_1| |z_2|$.

A3. Division (1 mark). Write each in the form $a + bi$, by multiplying top and bottom by the conjugate of the denominator:

- (a) $\frac{z_1}{z_2}$ (b) $\frac{1}{i}$ (c) $\frac{2+i}{1+i}$

A4. Polar form (1.5 marks). Write each number as $re^{i\theta}$ with $r > 0$ and $-\pi < \theta \leq \pi$. Draw each on the complex plane first.

- (a) $1 + i$ (b) $-\sqrt{3} + i$ (c) $-1 - i$ (d) $-3i$
- (e) For (b) and (c), compute $\arctan(b/a)$ on a calculator. Say in one sentence each why it is not the phase, and what you did instead.
- (f) Convert $2e^{i\pi/3}$ back to the form $a + bi$.

A5. Multiplication as rotation (1 mark). Let $z = 2 + i$.

- (a) Compute iz . Show that $|iz| = |z|$ and that the phase has increased by $\pi/2$. Mark z and iz on one sketch of the complex plane.
- (b) Find the complex number w such that multiplying any z by w rotates it 60° anticlockwise and leaves its modulus unchanged. Give w in polar form and in the form $a + bi$.

A6. Powers through the polar form (1.5 marks). Use $(re^{i\theta})^n = r^n e^{in\theta}$.

- (a) Compute $(1 + i)^8$.
- (b) Compute $(-1 + i)^6$. Check it a second way: square $-1 + i$ first, then cube the result.
- (c) Find both complex numbers z with $z^2 = i$. Give each in polar form and in the form $a + bi$.

A7. Euler's formula and phase factors (1 mark). Use $e^{i\theta} = \cos \theta + i \sin \theta$.

- (a) Write $e^{i\pi}$, $e^{i\pi/2}$, $e^{-i\pi/2}$ and $e^{2\pi i}$ in the form $a + bi$.

- (b) Show that $|e^{i\theta}| = 1$ for every real θ .
- (c) Hence show that $|e^{i\gamma}z| = |z|$ for any complex z and any real γ .

A8. Relative phase (1 mark). Let $z_1 = \frac{1}{\sqrt{2}}$ and $z_2 = \frac{1}{\sqrt{2}} e^{i\varphi}$, where φ is real. (These replace the z_1, z_2 above for this problem only.)

- (a) Show that $|z_1 + z_2|^2 = 1 + \cos \varphi$. Use $|w|^2 = w w^*$, and Euler's formula for $e^{i\varphi} + e^{-i\varphi}$.
- (b) Evaluate it for $\varphi = 0, \pi/2$ and π .
- (c) Now multiply **both** numbers by the same phase factor $e^{i\gamma}$. Show that $|e^{i\gamma}z_1 + e^{i\gamma}z_2|^2$ is unchanged.
- (d) In one sentence: which of the two phases, γ or φ , can change the result? Unit 3 comes back to this.

How to write the NumPy check file

Parts A and B each come with a short Python file that checks some of your hand answers with NumPy. Hand calculation comes first; NumPy is how you catch your own slips. The file must run from top to bottom without an error, and each check prints the problem number next to the value, so that the output can be read against your scan. Here is the format, on a number that is not in the problem set:

```
import numpy as np

z = -1 + 1j                                # in Python the imaginary unit is written 1j
print("example: |z| =", abs(z))            # by hand: sqrt(2) = 1.4142
print("example: arg z =", np.angle(z))     # by hand: 3*pi/4 = 2.3562
```

If NumPy disagrees with your hand answer, find out which is wrong and say so in a comment. Do not quietly change the hand answer to match: a comment such as `# my first attempt had the wrong quadrant; corrected in A4(c)` shows exactly the checking this file exists for.

NumPy check file (1 mark). `checks_a.py` checks at least **three** of your answers, including one from A4 and one from A6. Each printed line names its problem.

PROBLEM	A1	A2	A3	A4	A5	A6	A7	A8	NUMPY	TOTAL
Marks	1	1	1	1.5	1	1.5	1	1	1	10

Part B: linear algebra (14 marks)

Due at the end of **week 10**. Covers sessions 4 to 9. Submit `<ROLLN0>-partB.pdf` (the scan) and `<ROLLN0>-partB.py` (the NumPy checks).

The inner product conjugates the **first** argument: $\langle \vec{u}, \vec{v} \rangle = \vec{u}^\dagger \vec{v} = \sum_k u_k^* v_k$. Problems B8 and B9 use the two topics added beyond the syllabus in week 9, eigenvalues and the tensor product.

B1. Matrix products (1 mark). Let

$$A = \begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Compute $A\vec{v}$, AB and BA . Is $AB = BA$?

B2. The dagger of a product (1.5 marks). With A and B from B1, compute $A^\dagger$, $(AB)^\dagger$, $B^\dagger A^\dagger$ and $A^\dagger B^\dagger$. Which of the last two equals $(AB)^\dagger$? State the rule this illustrates.

B3. Inner products with complex entries (1.5 marks). Let $\vec{u} = (1, i)^T$ and $\vec{w} = (2, 1 + i)^T$.

- (a) Compute $\langle \vec{u}, \vec{w} \rangle$ and $\langle \vec{w}, \vec{u} \rangle$. How are they related?
- (b) Compute $\vec{u}^T \vec{w}$, the same sum with no conjugation, and compare.
- (c) Compute $\langle \vec{u}, \vec{u} \rangle$ and $\vec{u}^T \vec{u}$. Explain in one sentence why the inner product must conjugate one argument.

B4. Normalisation (1 mark).

- (a) Normalise $\vec{w} = (1 + i, 2, -i)^T$.
- (b) Find **every** complex number c for which $c(3, 4i)^T$ is a unit vector. (There is more than one.)

B5. Checking an orthonormal basis (1.5 marks). Let $\vec{b}_1 = \frac{1}{\sqrt{2}}(1, i)^T$, $\vec{b}_2 = \frac{1}{\sqrt{2}}(1, -i)^T$ and $\vec{c} = \frac{1}{\sqrt{2}}(i, -1)^T$.

- (a) Show that $\{\vec{b}_1, \vec{b}_2\}$ is an orthonormal basis of $\mathbb{C}^2$.
- (b) Compute $\langle \vec{b}_1, \vec{c} \rangle$, and also $\vec{b}_1^T \vec{c}$ with no conjugation. One of them says the two vectors are orthogonal. Which is right? Find the number k with $\vec{c} = k \vec{b}_1$ and use it to explain.
- (c) Is $\{(1, 1)^T, (1, -1)^T\}$ an orthonormal basis? If not, fix it.

B6. Components in another basis (1.5 marks). Let $\vec{e}_\pm = \frac{1}{\sqrt{2}}(1, \pm 1)^T$ and $\vec{v} = \frac{1}{\sqrt{3}}(1, 1 + i)^T$.

- (a) Check that $\vec{v}$ is a unit vector.
- (b) Compute the components $c_\pm = \langle \vec{e}_\pm, \vec{v} \rangle$ by projection.
- (c) Verify that $c_+ \vec{e}_+ + c_- \vec{e}_- = \vec{v}$.
- (d) Compute $|c_+|^2 + |c_-|^2$, and say why you expected that value.

B7. Hermitian and unitary (1 mark). Let

$$M = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}, \quad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

Decide, with working, whether M is Hermitian, whether U is unitary, whether U is Hermitian, and whether M is unitary.

B8. Eigenvalues and eigenvectors (2 marks). For the matrix M of B7:

- (a) Find its eigenvalues from $\det(M - \lambda I) = 0$.
- (b) Find a **normalised** eigenvector for each eigenvalue.
- (c) Check that the two eigenvectors are orthogonal. Which two facts about Hermitian matrices do (a) and (c) illustrate?

B9. The tensor product (2 marks).

- (a) Compute $(1, 2)^T \otimes (3, i)^T$ and $(3, i)^T \otimes (1, 2)^T$. Are they equal?
- (b) Compute the 4×4 matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
- (c) Show that if $\vec{w} = (p, q)^T \otimes (r, s)^T$ then $w_1 w_4 - w_2 w_3 = 0$. Hence show that $(1, 0, 0, 1)^T$ is **not** the tensor product of two two-entry vectors.
- (d) One of $(1, 2, 3, 6)^T$ and $(1, 2, 3, 4)^T$ is a tensor product of two two-entry vectors. Say which, and find the two factors.

NumPy check file (1 mark). `checks_b.py` checks at least **four** of your answers, including B8 and one part of B9. Useful calls: `A @ B` for a matrix product, `A.conj().T` for the dagger, `np.vdot(u, v)` for $\langle \vec{u}, \vec{v} \rangle$ (it conjugates the first argument, as we do), `np.linalg.eigh(M)` for a Hermitian eigenproblem, `np.kron(a, b)` for $\vec{a} \otimes \vec{b}$.

PROBLEM	B1	B2	B3	B4	B5	B6	B7	B8	B9	NUMPY	TOTAL
Marks	1	1.5	1.5	1	1.5	1.5	1	2	2	1	14

Part C: a qubit simulator in NumPy (20 marks)

Covers sessions 10 to 15. Accept the Classroom 50 item `mpqt-c` and commit `qubit.py` (your finished module) and `report.pdf` (the report).

The module (14 marks, autograded)

You complete `qubit.py`, a small state-vector simulator: kets as NumPy arrays, the inner product, Born probabilities, gates, measurement with collapse, tensor products and expectation values. The full specification, the starter file and a public tests are in your `mpqt-c` repository, and the full sheet is `exercises/mpqt-ex01-qubit-simulator` in the course repository; read it before you start. NumPy only: no qiskit or any other quantum library.

The grader runs your file through six stages. Run the public tests on your own machine before you submit, one stage at a time: `python tests/public.py qubit.py --stage states`, and so on for each stage.

STAGE	MARKS	WRITE IT AFTER
States and normalisation	2	Week 11
Inner product and Born rule	3	Week 12
Gates, unitarity and Hermiticity	3	Week 13
Measurement and collapse	2	Week 12
Tensor products and CNOT	2	Week 14
Expectation values	2	Week 15

Add a stage each week as Unit 3 is taught. Left to the final week, this is six functions and a report at once.

The report (6 marks)

Two to three pages, as a PDF. Run three experiments using **your own module**: import it and call its functions. Put the code for the experiments under `if __name__ == "__main__":` at the bottom of `qubit.py`, so the grader can import the file without running them. For each experiment give the few lines of code that produce the result, the seed you used, the numbers you got, and a short explanation.

Experiment 1: the Born rule (2 marks). Prepare $|+\rangle$ and measure it 1000 times in the computational basis $\{|0\rangle, |1\rangle\}$, then 1000 times in the basis $\{|+\rangle, |-\rangle\}$. Tabulate the counts of each outcome next to the counts the Born rule predicts. For 1000 shots with probability p , counts typically land within

about $\sqrt{1000 p(1-p)}$ of the prediction. Say whether yours do, and explain why one basis gives a random result and the other a certain one.

Experiment 2: interference (2 marks).

- (a) Show numerically that $H|0\rangle = |+\rangle$, $HH = I$ and $HZH = X$.
- (b) Start from $|0\rangle$ and apply H twice, then measure in the computational basis; repeat 1000 times. Then do the same with a measurement in the computational basis **between** the two H gates (use the collapsed state it returns). Tabulate both sets of counts.
- (c) Explain the difference by writing $H|+\rangle$ amplitude by amplitude: which contributions add and which cancel, and what the middle measurement does to that.

Experiment 3: entanglement (2 marks).

- (a) Prepare $|\Phi^+\rangle$ by applying $H \otimes I$ and then CNOT to $|00\rangle$. Print its four amplitudes.
- (b) Measure it 1000 times in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. Tabulate the counts and report the fraction of shots in which the two qubits gave the same bit. For comparison, do the same for $|+\rangle \otimes |+\rangle$.
- (c) Write the four amplitudes of a two-qubit state as a 2×2 array, $\begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}$. For a product state this array has $\psi_{00}\psi_{11} - \psi_{01}\psi_{10} = 0$ (Part B, problem B9). Compute this number for $|\Phi^+\rangle$ and for $|+\rangle \otimes |+\rangle$, and say what it shows about $|\Phi^+\rangle$.

PITFALL · REPORT WHAT YOU RAN

The counts in your report must be the ones your code prints with the seed you state. In the viva you may be asked to run it again. Numbers copied from the notes, or from someone else, are easy to spot: they are always a little too round.

Viva, submission and policies

The viva (6 marks)

Five minutes, one to one, in class after Part C closes; the slots are announced in class. Bring your laptop with your submitted `qubit.py` on it.

WHAT HAPPENS	MARKS
I point at a line of your <code>qubit.py</code> and you explain what it does and why it is there	3
I point at a step in your Part A or Part B scan and you explain it, or do a similar step on the spot	3

A missed viva, without documented cause, scores zero. If you cannot explain a part of your own submission, the marks for that part are reviewed as well.

What to submit

Everything is submitted through Classroom 50, classroom `qts-mpqt`. Each part is its own item, with its own private repository:

PART	ACCEPT	FILES IN THE REPOSITORY
A	<code>gh student accept Amrita-School-of-AI qts-mpqt mpqt-a</code>	<code>partA.pdf</code> , <code>checks_a.py</code>
B	<code>gh student accept Amrita-School-of-AI qts-mpqt mpqt-b</code>	<code>partB.pdf</code> , <code>checks_b.py</code>
C	<code>gh student accept Amrita-School-of-AI qts-mpqt mpqt-c</code>	<code>qubit.py</code> , <code>report.pdf</code>

The file names must be exactly these: the repository already says who you are, so no roll number goes in a name. Commit, then `gh student submit`. The Getting Started guide (`handbook/student-guide.pdf`) walks through every step, including the setup and a practice item, A0, that carries no marks.

Policies

Late. 80 percent within 48 hours of a part's deadline, then not accepted without documented cause. Each part is late or on time on its own.

Collaboration. Discuss ideas freely. Write your own working and your own code. Two scans with the same slip in the same line, or two modules with the same structure under different variable names, will both be asked about in the viva.

AI tools. Permitted for learning. You must be able to explain every line of your code and every step of your working in the viva; the viva is where that is checked. In this course a tool that does the arithmetic for you removes exactly the practice the marks are for, and the end-semester paper is written by hand.